

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: ToyWorld E-Commerce Database Management System

1.Database Design

1.1 Seller Table

Attribute	Key Type
Seller_ID	Primary Key
Seller_Name	NOT NULL
Contact_Number	-
Email	-
Address	-

1.2 Inventory Table

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Seller_ID	Foreign Key
Stock_Quantity	DEFAULT
Stock_Status	DEFAULT
Last_Updated	NOT NULL

2.Relationship

Seller → Inventory

- Seller.Seller_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Seller_ID → Foreign Key referencing Seller.Seller_ID
- Inventory.Product_ID → Foreign Key referencing Product.Product_ID
- Relationship: One-to-Many (1:M)
- One seller can have multiple inventory records.

3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID NUMBER(5) PRIMARY KEY,  
3     Seller_Name VARCHAR2(50) NOT NULL,  
4     Contact_Number VARCHAR2(15),  
5     Email VARCHAR2(50),  
6     Address VARCHAR2(100)  
7 );  
  
Table created.
```

The Seller table is created with Seller ID as the primary key. It stores the details of ToyWorld sellers.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID NUMBER(5) PRIMARY KEY,  
3     Product_ID NUMBER(5),  
4     Seller_ID NUMBER(5),  
5     Stock_Quantity NUMBER(5) DEFAULT 0,  
6     Stock_Status VARCHAR2(20) DEFAULT 'Available',  
7     Last_Updated DATE NOT NULL,  
8     CONSTRAINT fk_inventory_product  
9         FOREIGN KEY (Product_ID)  
10        REFERENCES Product(Product_ID),  
11     CONSTRAINT fk_inventory_seller  
12        FOREIGN KEY (Seller_ID)  
13        REFERENCES Seller(Seller_ID)  
14 );  
  
Table created.
```

The Inventory table is created with Inventory ID as the primary key. Seller ID and Product ID are used as foreign keys to establish relationships with the Seller and Product tables.

4. Data Insertion

4.1 Seller Records

Five seller records are inserted into the Seller table, followed by a COMMIT to save the transaction.

```

SQL> INSERT INTO Seller VALUES
  2 (101, 'Happy Toys', '9876543210', 'happytoys@gmail.com', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (102, 'Kids World', '9876501234', 'kidsworld@gmail.com', 'Coimbatore');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (103, 'Fun Zone Toys', '9876512345', 'funzone@gmail.com', 'Madurai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (104, 'Little Stars', '9876523456', 'littlestars@gmail.com', 'Salem');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (105, 'Toy Planet', '9876534567', 'toyplanet@gmail.com', 'Trichy');

1 row created.

```

4.2 Inventory Records

Inventory records are inserted with seller IDs, product IDs, stock quantities, stock status, and last updated dates.

```

SQL> INSERT INTO Inventory VALUES (301, 101, 101, 25, 'Available', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (302, 103, 102, 15, 'Available', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (303, 104, 103, 18, 'Available', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (304, 105, 104, 0, 'Out of Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (305, 108, 105, 28, 'Available', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (306, 109, 101, 0, 'Out of Stock', SYSDATE);

1 row created.

```

5.Data Retrieval

5.1 Display Seller Records

All seller records are retrieved to verify the successful insertion of seller information.

```
SQL> SELECT*FROM Seller;
```

SELLER_ID	SELLER_NAME	CONTACT_NUMBER	EMAIL
101	Happy Toys	9876543210	happytoys@gmail.com
102	Kids World	9876501234	kidsworld@gmail.com
103	Fun Zone Toys	9876512345	funzone@gmail.com
104	Little Stars	9876523456	littlestars@gmail.com
105	Toy Planet	9876534567	toyplanet@gmail.com

5.2 Display Inventory Records

All inventory records are retrieved to verify the stock information maintained in the database.

```
SQL> SELECT*FROM Inventory;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
301	101	101	25	Available	17-SEP-26
302	103	102	15	Available	17-SEP-26
303	104	103	18	Available	17-SEP-26
304	105	104	0	Out of Stock	17-SEP-26
305	108	105	28	Available	17-SEP-26
306	109	101	0	Out of Stock	17-SEP-26

6 rows selected.

6. Seller Product Information

A JOIN query is used to display the seller name, product name, product ID, stock quantity, stock status, and last updated date.

```

SQL> SELECT
2     s.Seller_Name,
3     i.Product_ID,
4     i.Stock_Quantity,
5     i.Stock_Status,
6     i.Last_Updated
7 FROM Seller s
8 JOIN Inventory i
9 ON s.Seller_ID = i.Seller_ID;

```

SELLER_NAME	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
Happy Toys	101	25	Available	17-SEP-26
Kids World	103	15	Available	17-SEP-26
Fun Zone Toys	104	18	Available	17-SEP-26
Little Stars	105	0	Out of Stock	17-SEP-26
Toy Planet	108	28	Available	17-SEP-26
Happy Toys	109	0	Out of Stock	17-SEP-26

6 rows selected.

7. Product Availability

7.1 Available Products

Products with available stock are displayed using the Stock_Status field.

```

SQL> SELECT
2     p.Product_ID,
3     p.Product_Name,
4     i.Stock_Quantity,
5     i.Stock_Status
6 FROM Product p
7 JOIN Inventory i
8 ON p.Product_ID = i.Product_ID
9 WHERE i.Stock_Status = 'Available';

```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
101	ABC Learning Kit	25	Available
103	Remote Car	15	Available
104	Chess Board	18	Available
108	Jigsaw Puzzle	28	Available

7.2 Unavailable Products

Products with zero stock or Out of Stock status are displayed separately.

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM Product p
  7 JOIN Inventory i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Stock_Status = 'Out of Stock'
 10      OR i.Stock_Quantity = 0;
```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
105	Football	0	Out of Stock
109	Superhero Figure	0	Out of Stock

8. Inventory Update

Stock quantity, stock status, and last updated date are updated to maintain the latest inventory information.

```
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 20,
  3     Stock_Status = 'Available',
  4     Last_Updated = SYSDATE
  5 WHERE Inventory_ID = 304;

1 row updated.
```

```
SQL> SELECT*FROM Inventory;

INVENTORY_ID PRODUCT_ID SELLER_ID STOCK_QUANTITY STOCK_STATUS LAST_UPDA
-----
          301         101         101           25 Available 17-SEP-26
          302         103         102           15 Available 17-SEP-26
          303         104         103           18 Available 17-SEP-26
          304         105         104           20 Available 17-SEP-26
          305         108         105           28 Available 17-SEP-26
          306         109         101              0 Out of Stock 17-SEP-26

6 rows selected.
```

9. Inventory Status Reports

9.1 Complete Inventory Report

```

SQL> SELECT
  2     i.Inventory_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status,
  7     i.Last_Updated
  8 FROM Inventory i
  9 JOIN Product p
10 ON i.Product_ID = p.Product_ID
11 JOIN Seller s
12 ON i.Seller_ID = s.Seller_ID;

```

INVENTORY_ID	PRODUCT_NAME	SELLER_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
301	ABC Learning Kit	Happy Toys	25	Available	17-SEP-26
302	Remote Car	Kids World	15	Available	17-SEP-26
303	Chess Board	Fun Zone Toys	18	Available	17-SEP-26
304	Football	Little Stars	20	Available	17-SEP-26
305	Jigsaw Puzzle	Toy Planet	28	Available	17-SEP-26
306	Superhero Figure	Happy Toys	0	Out of Stock	17-SEP-26

6 rows selected.

9.2 Total Stock by Seller

```

SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Stock_Quantity) AS Total_Stock
  4 FROM Seller s
  5 JOIN Inventory i
  6 ON s.Seller_ID = i.Seller_ID
  7 GROUP BY s.Seller_Name;

```

SELLER_NAME	TOTAL_STOCK
Happy Toys	25
Kids World	15
Fun Zone Toys	18
Little Stars	20
Toy Planet	28

9.3 Product with Zero Stock

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock_Quantity
  5 FROM Product p
  6 JOIN Inventory i
  7 ON p.Product_ID = i.Product_ID
  8 WHERE i.Stock_Quantity = 0;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
          109
Superhero Figure
          0
```

9.4 Available and Unavailable Product Count

```
SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Product_Count
  4 FROM Inventory
  5 GROUP BY Stock_Status;
```

STOCK_STATUS	PRODUCT_COUNT
Available	5
Out of Stock	1

Conclusion

The ToyWorld E-Commerce Database Management System successfully manages product, seller, and inventory information using SQL. The system uses primary keys and foreign keys to establish relationships between the tables and maintains accurate stock details. SQL JOIN, INSERT, UPDATE, DELETE, and reporting queries are used to manage seller products, track inventory, and identify available and out-of-stock products.